

Introduction

Computer Network Technologies and
Services

Claudio Casetti, Marco Rapelli



Teachers

- **Prof. Claudio Casetti**

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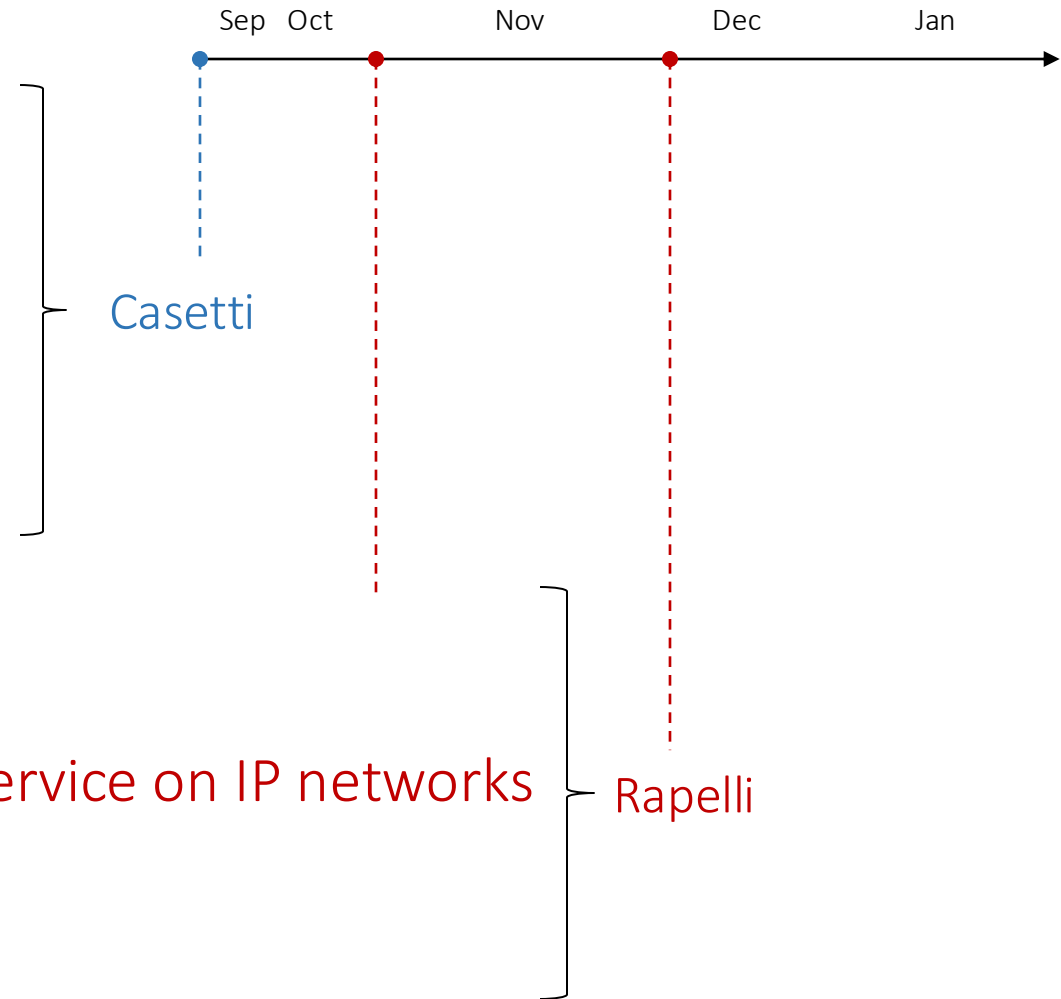
- **Prof. Marco Rapelli**

- Dipartimento di Automatica e Informatica
- Email: marco.rapelli@polito.it



Topics

- IPv4 network design
- IPv6 protocol and network design
- Cellular networks
- Ethernet local area network design
- Principles of dynamic routing
- Virtual Private Networks (VPN)
- MPLS protocol architecture
- Protocols and Techniques for quality of service on IP networks
- Optical networks
- Traffic Pattern Analysis



Expected knowledge

- To proficiently follow this course, you must be conversant in the following topics, taught in BSc courses
 - Architecture of a network: what is a node, a link, a channel
 - Packet switching: what is a packet, its structure (header+data), what “switching” means
 - Protocols: what is a protocol, what are the main protocols in use today
 - Layered Architecture: what is a network layer, the 7 OSI layers, their functions and interactions
 - Layer 1 and Layer 2 functions: they are not taught here, but you must be familiar with them
 - The course will focus on layers 3 and above, but you must know their basic functions and protocols
- If this is not the case, send me an email, I will provide you with a link to our BSc course material (to be reviewed on your own)



Schedule 2025-26

- Monday: 8:30-10:00 room 5
- Thursday: 16:00-19:00 room 4P
- Friday: 13:00-16:00 room 3P

Note: Until 17/10, the Friday class will be from 14:30 to 16:00



Labs

- Three lab experiences
 - IPv6 configuration
 - VPN configuration
 - Traffic Pattern Analysis
- Virtual labs (Crownlabs) – in classroom
 - 4 teams
 - Duration: 3h timeslots



Schedule 2025-26

Week	MO 8:30-10:00	TH 16:00 – 19:00	FR 13:00-16:00
22/09/25	CC	CC	CC
29/09/25	CC	CC	CC
06/10/25	CC	Lab IPv6 - 1	CC
13/10/25	CC	Lab IPv6 - 2	CC
20/10/25	No class	MR	MR
27/10/25	CC	Lab IPv6 - 3	CC
03/11/25	No class	Lab VPN - 1	Lab IPv6 - 4
10/11/25	CC	Lab VPN - 2	CC
17/11/25	CC	Lab VPN - 3	CC
24/11/25	CC	MR	Lab VPN - 4
01/12/25	MR	MR	MR
08/12/25	MR	Lab Traffic - 1	MR
15/12/25	MR	Lab Traffic - 2	MR
07/01/26	MR	Lab Traffic - 3	Lab Traffic - 4

14:30-16:00



Material

- Polito e-learning portal
 - Slides
 - Recorded lectures



Exam

- **Mandatory:** 1-hour written exam, including 33 multiple-choice questions
 - Correct answer: 1 point
 - Wrong answer: -1/3 points
 - No answer: 0 points
 - Final mark rounded to nearest integer (25 and 1/3 -> 25; 25 and 2/3 -> 26)
- **No Oral Exam**



One more thing... Rules of Communication

- When you write to any of the Professors always state:
 - Your name and matricola number
 - The title of the course (each of us teaches multiple courses)
 - E.g., “I am in your Networks course” is not enough

This will speed up our reply!
- When you register for an exam and cannot attend: unregister or send an email
 - This will allow us to optimize the logistics and to avoid printing additional exam material
- Your exam mark is **final**:
 - Emails asking for:
 - a redress of your mark (e.g., “please add 1.6 points so I can pass, otherwise I lose my scholarship”)
 - an oral exam
 - an extra exam session
 - registration of an exam in a different session
 - ...

will not be answered

